

Seq2Seq and Embedding-based Mini-Translators

Joint Final Project NLP and DL - Proposal

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Introduction

The goal of this project is to build a translation model that is capable of translating between two languages. The model will be based on the Seq2Seq architecture, which consists of two main components: an encoder and a decoder. The former takes an input sequence and encodes it into a fixed-size representation, while the latter takes this representation and decodes it into an output sequence. This structure will be later explained in more detail.

Moreover, the model will be trained using embeddings to represent the words in the input and output languages. This will allow the model to treat words as vectors in a continuous space, which can help capture the semantic relationships between them. This fulfills the requirements included in the given project description.

In order to evaluate the available options to build the model, we have conducted research to understand different approaches and techniques that can be valuable to build a translation model. Some of these include, among others, the use of attention mechanisms and transformers, albeit these will be further explored in the next sections.

The project will be implemented in Python using the PyTorch library. Once the model is built, it will be trained on a dataset of parallel sentences in two languages. Subsequently, the model will be evaluated on a separate test set to assess its performance. Based on the results, we will analyze the strengths and weaknesses of the model and propose potential improvements.

All the code and documentation will be available (and regularly updated) on a GitHub repository, which can be accessed *here*.

Conducted Research

As aforementioned, we have investigated several approaches by reading papers and articles related to the goal of the project. The two following are the most relevant ones found so far.

Paper 1 Exploiting Similarities among Languages for Machine Translation

This paper describes a method that allows translations between languages, regardless of whether these are similar or not. Firstly, embeddings are used to represent words as vectors, in which each of its dimensions represents a different attribute of the word. In this way, similar words appear closer in the vector space, which allows the model to capture the semantic relationships between them, and linear transformations can be applied to the word vectors. Since the embeddings are distributed in a similar way in both languages, a translation matrix can be designed to map the embeddings of one language to the other. Two model types are proposed: on the one hand, the CBOW model (Continuous Bag of Words) predicts a word given its context, i.e., its surrounding words; and on the other hand, the Skip-gram model predicts which words are likely to appear in the context of a given word by trying to maximize the average of the log probability. Both of them are trained using backpropagation and stochastic gradient descent, and are able to recognize the semantic implications among the words. Given its low complexity, they can be trained very quickly. A relevant piece of advice found in this paper is to carefully identify the most accurate way to preprocess the data, as it can have a significant impact on the model's performance.

Paper 2 Neural Machine Translation: Inner Workings, Seq2Seq, and Transformers This second paper brings to light important facts about models that may result useful for the translator.

- Sequence to sequence (Seq2Seq) model: it is composed of two main components, an encoder and a decoder. Each of them consists of a recurrent neural network (RNN).
- Attention mechanism: it allows the model to focus on different parts of the input sequence when generating the output sequence. It reduces the number of hidden states that the decoder receives by applying dot products.

- Transformers: they are a type of model that uses self-attention to process sequences of data. They are more efficient than RNNs and can capture long-range dependencies in the data. They also include an encoder and a decoder.

Project Plan

Our initial approach for this project will be now briefly described. Notwithstanding, changes in this idea may occur as we develop the model during the following weeks. Considering we aim to create the best possible version of the translator, we are willing to continue researching and to apply other techniques that can lead us to a better outcome. It goes without saying that any advice or suggestions provided by the professors will be greatly appreciated.

A. Model Idea

Based on the performed research, we have decided to build our translator using a Transformer model (besides the embeddings and Seq2Seq architecture). Owing to its structure, the training of its weights is not sequential but parallel, which proves to be a significant advantage if multiple cores (GPUs) are available. Another key reason that supports this choice is that it allows the application of the attention mechanism (both self-attention and encoder-decoder attention), apart from the fact that this model is easily comprehended by our team members.

B. Dataset

In the natural language processing field, the development of automatic translation models requires a robust multilingual dataset to effectively train the algorithms. In this project, the dataset *here* attached will be used, as it incorporates a wide variety of texts and phrases in different languages. These files, after its respective preprocessing, will be crucial for the correct training of the model and for its future fine-tuning.

C. Evaluation

To evaluate our model, we have chosen to assess its accuracy by observing the proportion of correctly translated words and thereby rate the quality of the model. In addition, the CER (Character Error Rate) metric will be explored to determine if its use is suitable (and profitable) for our model. All the same, it is worth bearing in mind the importance of verifying the steps taken in the construction of the model. For instance, in the embedding layer it should be checked that the obtained numeric vectors are representative, and that the embedding translation between languages is correctly applied.

Milestones

12th April Aiming at the first project report, the idea is to implement a basic model trained in a small dataset of two languages. Initially, the languages taken into consideration will share the same alphabet and will preferably originate from the same language family (for instance, romance languages) so that they have similar characteristics. After studying the outcome of this first approximation, we will consider improvements and further steps to develop a better version of the model.

19th April By the time of the submission of the second progress report, we expect to have several training attempts of a more advanced model, using the simpler languages in the dataset. Once this is fulfilled, we intend to implement the translation of more complex languages, and optimize the model based on the results obtained.

Final Paper Provided that the previous weeks have been successful and that the project is achieved beforehand, we will seek to develop a series of improvements (which will be adjusted according to the remaining time and the workload they entail) which may consist of, for example, a visual interface to interact with the created translators or an automatic language detection system.